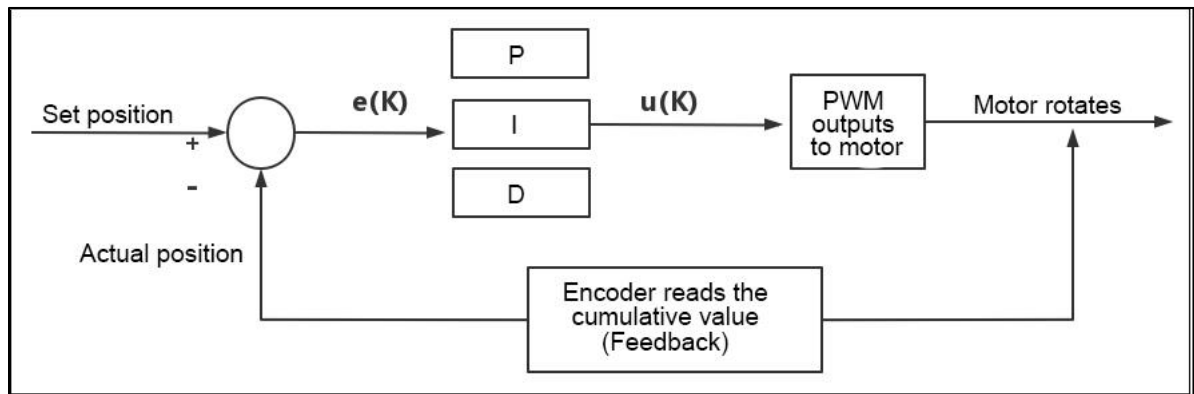


Position PID Control Basics

Position closed-loop control means that the position information of the motor is measured based on the pulse accumulation of the encoder and compared with the target value to obtain the control deviation. Then the process of controlling the proportional (P), integral (I) and differential (D) of the deviation to make the deviation tend to zero.



You can refer to the following formulate:

$$\text{formula: } \text{PWM} = K_p e(k) + K_i \sum e(k) + K_d [e(k) - e(k-1)]$$

$e(k)$: current deviation

$\sum e(k)$: deviation accumulation

$e(k-1)$: previous deviation

K_p : proportionality coefficient

K_i : Integral coefficient

K_d : differential coefficient